A More Details of the Dataset

A.1 Dataset Overview for Responsible NLP

Documentation of the artifacts:.

- Coverage of domains: general domain (e.g., educational examples of logical fallacies), and climate change news articles with logical fallacies.

- Languages: English.

- Linguistic phenomena: Logical fallacies.

- Demographic groups represented: No specific demographic groups.

Annotation details:.

- Basic demographic and geographic characteristics of the annotator population that is the source of the data: All annotators are native English speakers who are undergraduates at a university in the US. There are two male annotators and two female annotators.

- How you recruited (e.g., crowdsourcing platform, students) and paid participants, and discuss if such payment is adequate given the participants' demographic (e.g., country of residence): We broadcast the recruitment to the undergraduate CS student mailing list at a university. We received a large number of applications and selected four annotators. We followed the university's standard payment of 14 USD/hour for each student.

- How consent was obtained from annotators: We explained to the annotators that the data will be open-sourced for research purpose.

- Data collection protocol approved (or determined exempt) by an ethics review board: The dataset included in this work did not go through reviews by an ethics review board.

- Full text of instructions given to participants: We first show to the participants the description and examples of the 13 logical fallacy types as in Appendix D, and when they are actually annotating, the interface screenshots are in Figures 3 and 4.

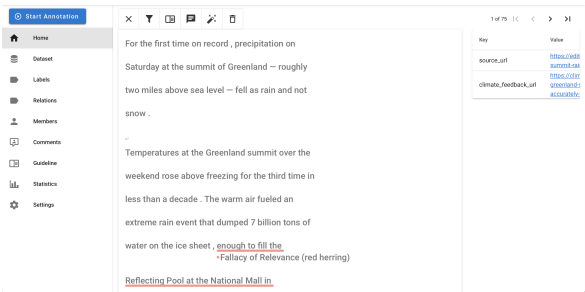

Figure 3: Annotation interface for the LOGICCLIMATE challenge set.

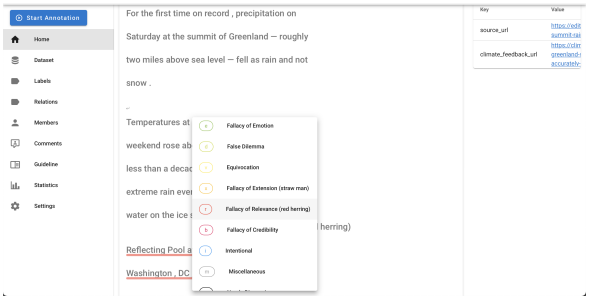

Figure 4: Choices of logical fallacy types in the annotation interface of the LOGICCLIMATE challenge set.

Data sheet:.

- Why was the dataset created: We created the dataset for the proposed logical fallacy classification task.

- Who funded the creation of the dataset: The LOGIC part was collected by the co-authors, and the LOGICCLIMATE part was collected using the funding of a professor at the university.

- What preprocessing/cleaning was done: We tokenized the text using the word tokenization function of NLTK.[4]

- Will the dataset be updated; how often, by whom: No, the dataset will be fixed.

Additional ethical concerns:.

- Whether the data that was collected/used contains any information that names or uniquely identifies individual people or offensive content: No, the dataset does not contain personal information.

- License or terms for use and/or distribution: The dataset is open-sourced with the MIT license, and the intended use is for academic research but not commercial purposes.

A.2 Data Filtering Details of LOGIC

The data automatically crawled from quiz websites contain lots of noises, so we conducted multiple filtering steps. The raw crawling by keyword matching such as "logic" and "fallacy" gives us 52K raw, unclean data samples, from which we filtered to 1.7K clean samples.

As not all of the automatically retrieved quizzes are in the form of "Identify the logical fallacy in this example: [...]", we remove all instances where the quiz question asks about irrelevant things such as the definition of a logical fallacies, or quiz questions with the keyword "logic" but in the context of other subjects such as logic circuits for electrical engineering, or pure math logic questions. This is

[4] https://nltk.org/

Fallacy Name	Description	Logical Form
Faulty Generalization	An informal fallacy wherein a conclusion is drawn about all or many instances of a phenomenon on the basis of one or a few instances of that phenomenon. is an example of jumping to conclusions.	[MSK1] has attribute [MSK2]. [MSK1] is a subset of [MSK3]. Therefore, all [MSK3] has attribute [MSK2]. (Reference)
False Causality	A statement that jumps to a conclusion implying a causal relationship without supporting evidence	[MSK1] occurred, then [MSK2] occurred. Therefore, [MSK1] caused [MSK2]. (Reference)
Circular Claim	A fallacy where the end of an argument comes back to the beginning without having proven itself.	[MSK1] is true because of [MSK2]. [MSK2] is true because of [MSK1]. (Reference)
Ad Populum	A fallacious argument which is based on affirming that something is real or better because the majority thinks so.	A lot of people believe [MSK1]. Therefore, [MSK1] must be true. (Reference)
Ad Hominem	An irrelevant attack towards the person or some aspect of the person who is making the argument, instead of addressing the argument or position directly.	[MSK1] is claiming [MSK2]. [MSK1] is a moron. Therefore, [MSK2] is not true. (Reference)
Deductive Fallacy	An error in the logical structure of an argument.	If [MSK1] is true, then [MSK2] is true. [MSK2] is true. Therefore, [MSK1] is true. (Reference)
Appeal to Emotion	Manipulation of the recipient's emotions in order to win an argument.	[MSK1] is made without evidence. In place of evidence, emotion is used to convince the interlocutor that [MSK1] is true. (Reference)
False Dilemma	A claim presenting only two options or sides when there are many options or sides.	Either [MSK1] or [MSK2] is true. (Reference)
Equivocation	An argument which uses a key term or phrase in an ambiguous way, with one meaning in one portion of the argument and then another meaning in another portion of the argument.	[MSK1] is used to mean [MSK2] in the premise. [MSK1] is used to mean [MSK3] in the conclusion. (Reference)
Fallacy of Extension	An arugment that attacks an exaggerated or caricatured version of your opponent's position.	[MSK1] makes claim [MSK2]. [MSK3] restates [MSK2] (in a distorted way). [MSK3] attacks the distorted version of [MSK2]. Therefore, [MSK2] is false. (Reference)
Fallacy of Relevance	Also known as red herring, this fallacy occurs when the speaker attempts to divert attention from the primary argument by offering a point that does not suffice as counterpoint/supporting evidence (even if it is true).	It is claimed that [MSK1] implies [MSK2], whereas [MSK1] is unrelated to [MSK2] (Reference)
Fallacy of Credibility	An appeal is made to some form of ethics, authority, or credibility.	[MSK1] claims that [MSK2]. [MSK1] are experts in the field concerning [MSK2]. Therefore, [MSK2] should be believed. (Reference)
Intentional Fallacy	A custom category for when an argument has some element that shows intent of a speaker to win an argument without actual supporting evidence.	[MSK1] knows [MSK2] is incorrect. [MSK1] still claim that [MSK2] is correct using an incorrect argument.

Table 9: Types of logical fallacies along with their descriptions and logical forms.

done by writing several matching patterns. After several processing steps such as deleting duplicates, we end up with 7,389 quiz questions. Moreover, as there is some noise that cannot be easily filtered by pattern matching, we also manually go through the entire dataset to only keep sentences that contain examples of logical fallacies, but not other types of quizzes.

The entire cleaning process resulted in 1.7K high-quality logically fallacious claims in our dataset. As a reference, for each fallacy example we also release the URL of the source website where we extract this example from.

A.3 Logical Fallacy Types

As different sources use different names for logical fallacies, we composed a set of 13 logical fallacy categories by conforming to set of logical fallacies given by Wikipedia,[5] and considering the most common types in the dataset. Therefore, we merged different surface forms of the same logical fallacy by listing out the different names of the same logical fallacy introduced on Wikipedia and also provided by educational websites. This leads to a reduction in logical fallacy types. For example, "hasty induction" and "jumping to conclusions" are merged under the category of "hasty generalization." We further improve the eventual list by handcrafted rules, and delete data samples that cannot be matched to any of the logical fallacy types in our list. For a small number of remaining logical fallacy names which we cannot resolve au-

[5] https://en.wikipedia.org/wiki/List_of_fallacies

tomatically, we ask human annotators to align the names.

We introduce the detailed description for each of the 13 logical fallacy types and their logical forms in Table 9. Most the descriptions and logical forms are collected from online websites introducing these logical fallacies. We provide the link of the source websites as references, and in some cases, we paraphrase the logical form to make it closer to natural text. In addition, we also provide the introduction of the 13 types that we compiled for the annotators in Appendix D.

A.4 Data Annotation Details of LOGICCLIMATE

To ensure good annotation quality, we ask the annotators to pass a test batch of 65 samples after reading the definitions and examples of the 13 logical fallacies in the LOGIC dataset carefully. The test batch consists of 5 randomly selected samples for each of the 13 logical fallacies, and the annotators achieve above 85% accuracy. We explained the examples where they did incorrectly and resolved their questions before they started annotating the LOGICCLIMATE test set.

Since each sample is annotated by two different annotators, we finalize the ground-truth labels in the following way: The two annotators merge all their annotations, and for places with divergent opinions, they cross-check with the experts' written reviews on the Climate Feedback website for each article. Specifically, each article is commented on by multiple expert reviewers such as professors, senior scientists and other researchers who explain what is fallacious with the article. If the labels can still not be unified after checking the expert reviews, since the annotators are trained to master the tasks very well (with 5+ hours of training, testing and discussions before the annotation), we let the two annotators have a discussion to decide the final label.

A.5 LOGICCLIMATE Examples

We also show examples of LOGICCLIMATE in Table 10.

B Implementation Details

B.1 Details of Zero-Shot Baselines

For the zero-shot classification models, we used pretrained NLI models (Yin et al., 2019) that are default choices of zero-shot classifiers in the transformers library (Wolf et al., 2020): BART-MNLI and RoBERTa-MNLI. For the implementation, we follow the standard pipeline introduced by huggingface.[6]

For the TARS model (Halder et al., 2020b), we follow the official documentation.[7] All the above zero-shot classifiers show reasonable performance on existing datasets.[8]

For GPT-3, we follow the official guide,[9] and use the following prompt (without additional efforts on prompt tuning because we do not assume any training samples for the zero-shot classification model): "Please classify a piece of text into the following categories of logical fallacies: [a list of all logical fallacy types].

Text: [Input text]

Label:"

We use the default search engine "davinci," and the model "curie." For reproducibility, we set the temperature to 0 for GPT-3 and all our zero-shot classification codes use a random seed of 1.

B.2 Details of Finetuned Models

All models are finetuned using the NLI task, as motivated in Section 3.1. We used a learning rate of 2^{-5}, and the AdamW optimizer. We do not turn the learning rate and optimizers since the loss converges smoothly in all cases. We tune the weights of the positive class, and set it to be 12, and the negative classes to be 1. The models used in our experiments have between 11M and 140M parameters. We train all models using NVIDIA TITAN RTX machines for less than two GPU Hours. For reproducibility, we fix the random seed to zero, and report the statistics of a single run.

Due to the different dataset nature, our LOGIC is a single-label multi-class classification and the LOGICCLIMATE is a multi-label multi-class classification.

B.3 Details of Our Structure-Aware Model

For the structure-aware classifier, we set the threshold of cosine similarity between two text spans to be 0.7, which is tuned using a manual grid search

[6] https://bit.ly/3E92Mvq

[7] https://github.com/flairNLP/flair/blob/master/resources/docs/TUTORIAL_10_TRAINING_ZERO_SHOT_MODEL.md

[8] https://github.com/nlptown/nlp-notebooks/blob/master/Zero-Shot%20Text%20Classification.ipynb

[9] https://beta.openai.com/docs/api-reference/classifications